VICTORY 2027

Natural Vegetation

LECTURE-3 Geography

Sameer Sir



Topics To Be Covered

1. Types of Vegetation
2. Forest conservation
3.
4.

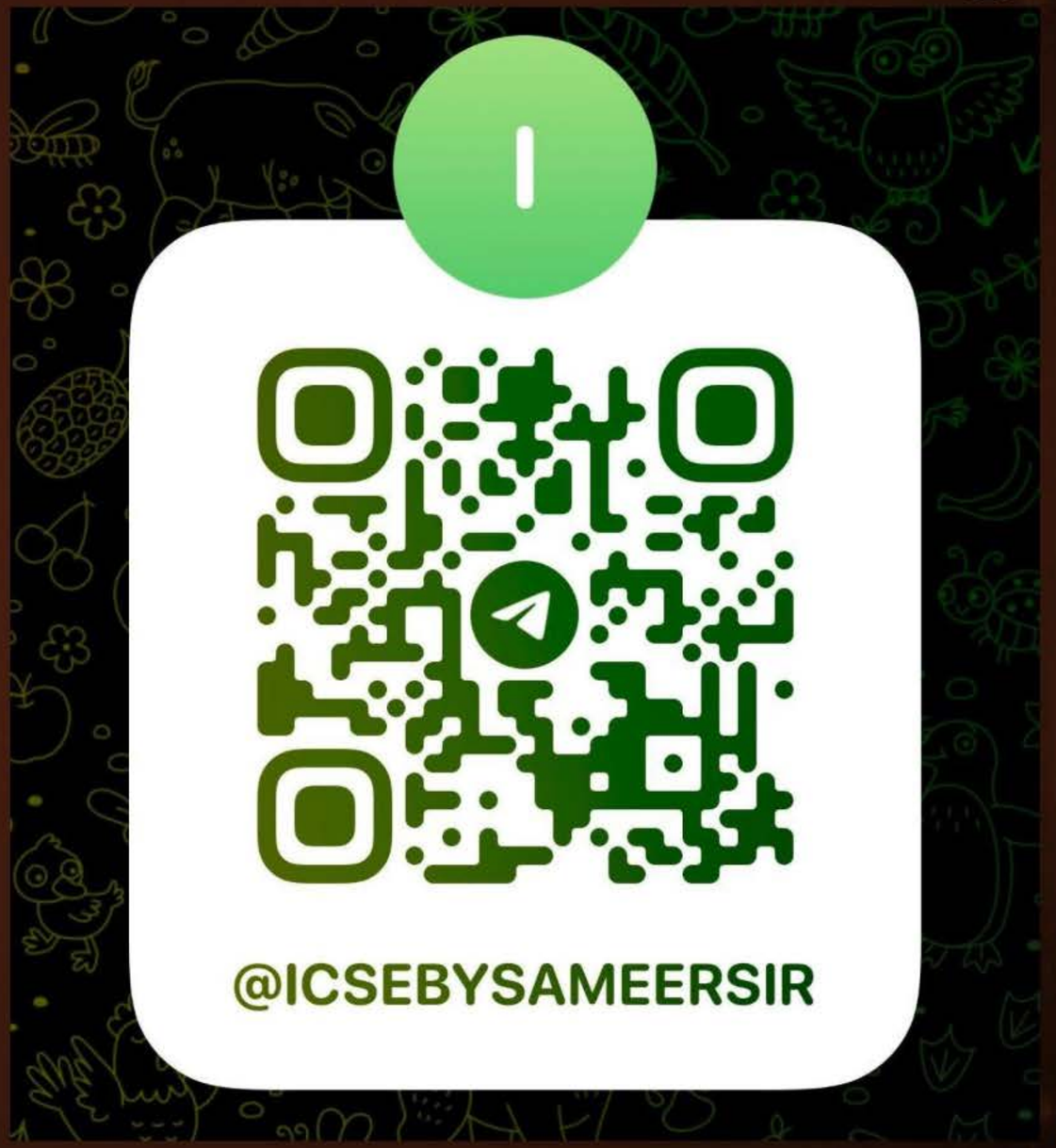
Natural Vegetation

- *Importance of forests.*
- *Types of vegetation; distribution and correlation with their environment (tropical evergreen, tropical deciduous, tropical desert, littoral and mountain).*
- *Forest conservation – afforestation, reforestation, social forestry, agroforestry and Van Mahotsav*


Goku is
here
Lets Start

Telegram

https://t.me/icsebysameersir



[ICSE Pattern] Give reason: "Trees in tropical desert regions have thick, waxy, or fleshy stems."

- A) To shed leaves twice a year
- B) To support dense climbing vines
- C) To allow rapid transpiration in sunlight
- D) To store water and reduce evaporation losses

Tropical Evergreen trees do not shed their leaves simultaneously because:

- A) The climate remains warm throughout the year.
- B) They receive heavy rainfall only during winter.
- C) Different species shed leaves at different times of the year.
- D) They have needle-like leaves to prevent transpiration.

Reason ?

Breathing roots (Pneumatophores) are a adaptation feature of trees found in:

- A) Tropical Deciduous Forests
- B) Montane Forests
- C) Tidal/Mangrove Forests
- D) Thorn and Scrub Forests

The most widespread and economically most important forests in India are:

- A) Tropical Deciduous (Monsoon) Forests
- B) Tropical Evergreen Forests
- C) Tidal Forests
- D) Thorn and Scrub Forests

In Thorn and Scrub vegetation, plants have long taproots primarily to:

- A) Store excess starch.
- B) Reach deep groundwater reserves.
- C) Support heavy branches.
- D) Absorb atmospheric moisture directly.

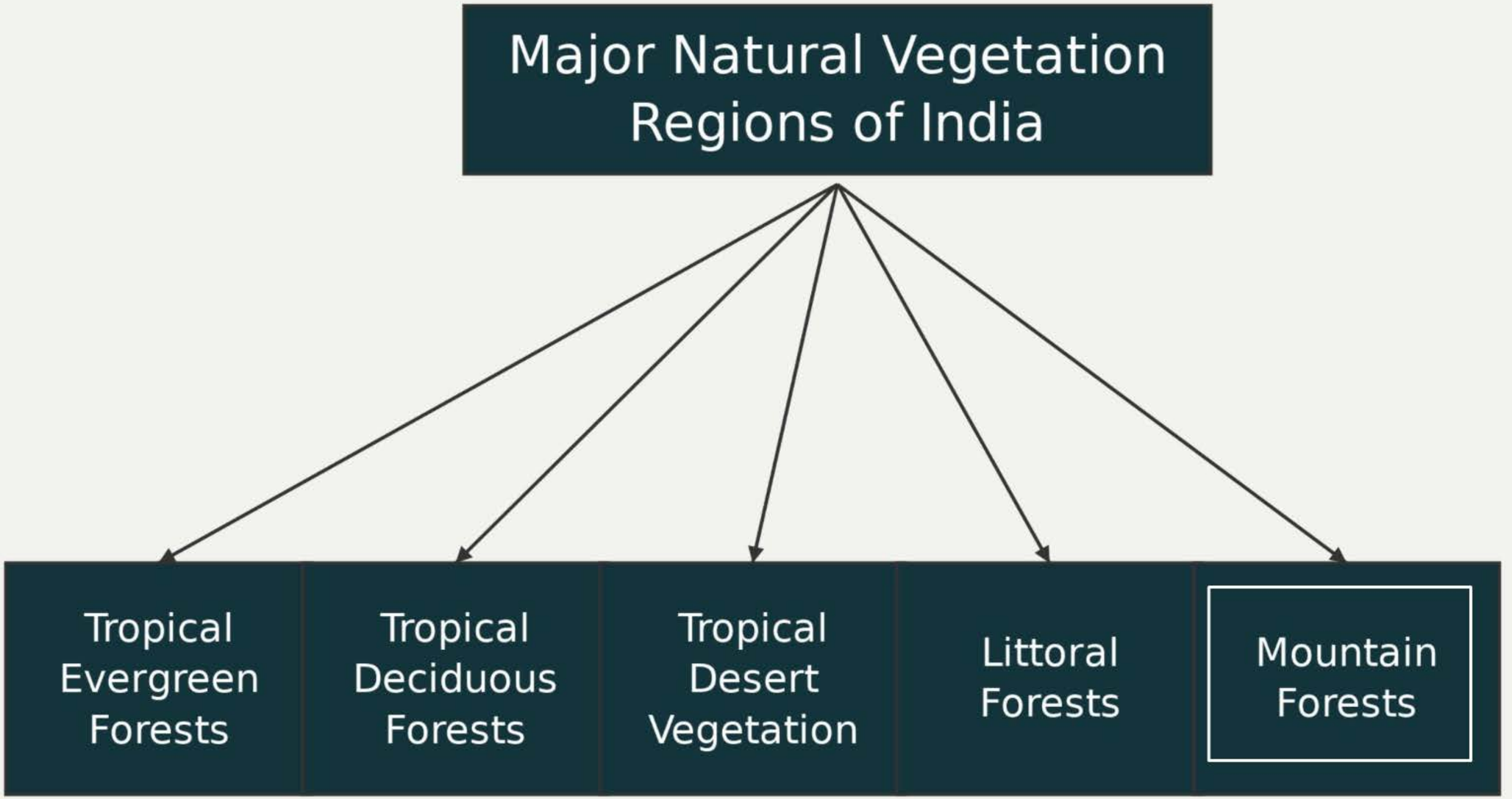
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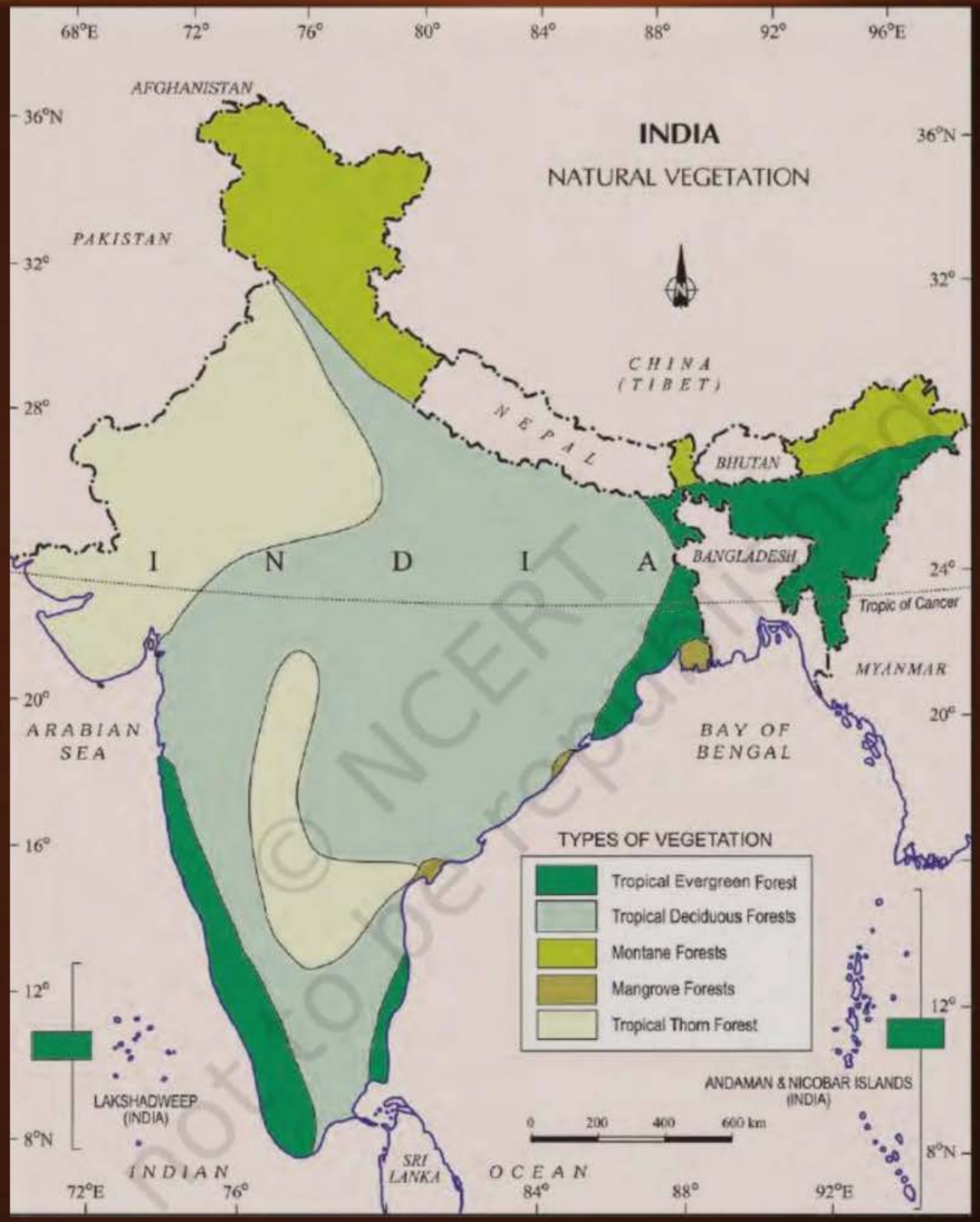
PYQ

[ICSE Pattern] Identify the forest type based on the given climatic conditions:

- *Temperature:* 24°C to 27°C | *Rainfall:* 100 cm to 200 cm | *Humidity:* Moderate
- A) Desert Vegetation
- B) Alpine Meadow
- C) Tidal Forest
- D) Tropical Deciduous Forest


Himalaya
6000m
1500m
Deciduous Forest
Peninsula
Shola


Major Natural Vegetation Regions of India
Tropical Evergreen Forests
Tropical Deciduous Forests
Tropical Desert Vegetation
Littoral Forests
Mountain Forests


INDIA
NATURAL VEGETATION
AFGHANISTAN
PAKISTAN
CHINA (TIBET)
NEPAL
BHUTAN
BANGLADESH
MYANMAR
I N D I A
Tropic of Cancer
ARABIAN SEA
BAY OF BENGAL
INDIAN OCEAN
SRI LANKA
LAKSHADWEEP (INDIA)
ANDAMAN & NICOBAR ISLANDS (INDIA)
TYPES OF VEGETATION
Tropical Evergreen Forest
Tropical Deciduous Forests
Montane Forests
Mangrove Forests
Tropical Thorn Forest
0 200 400 600 km
68°E
72°
76°
80°
84°
88°
92°
96°E
36°N
32°
28°
24°
20°
16°
12°
8°N

Type of Vegetation → Climate
→ Type of Soil
→ Topography

Normal lapse rate
Altitude
→ 165
166 m → 1°c ↓

Mountain Forest
Type of vegetation changes with altitude.
Snowfall

5. Mountain Forests

Montane

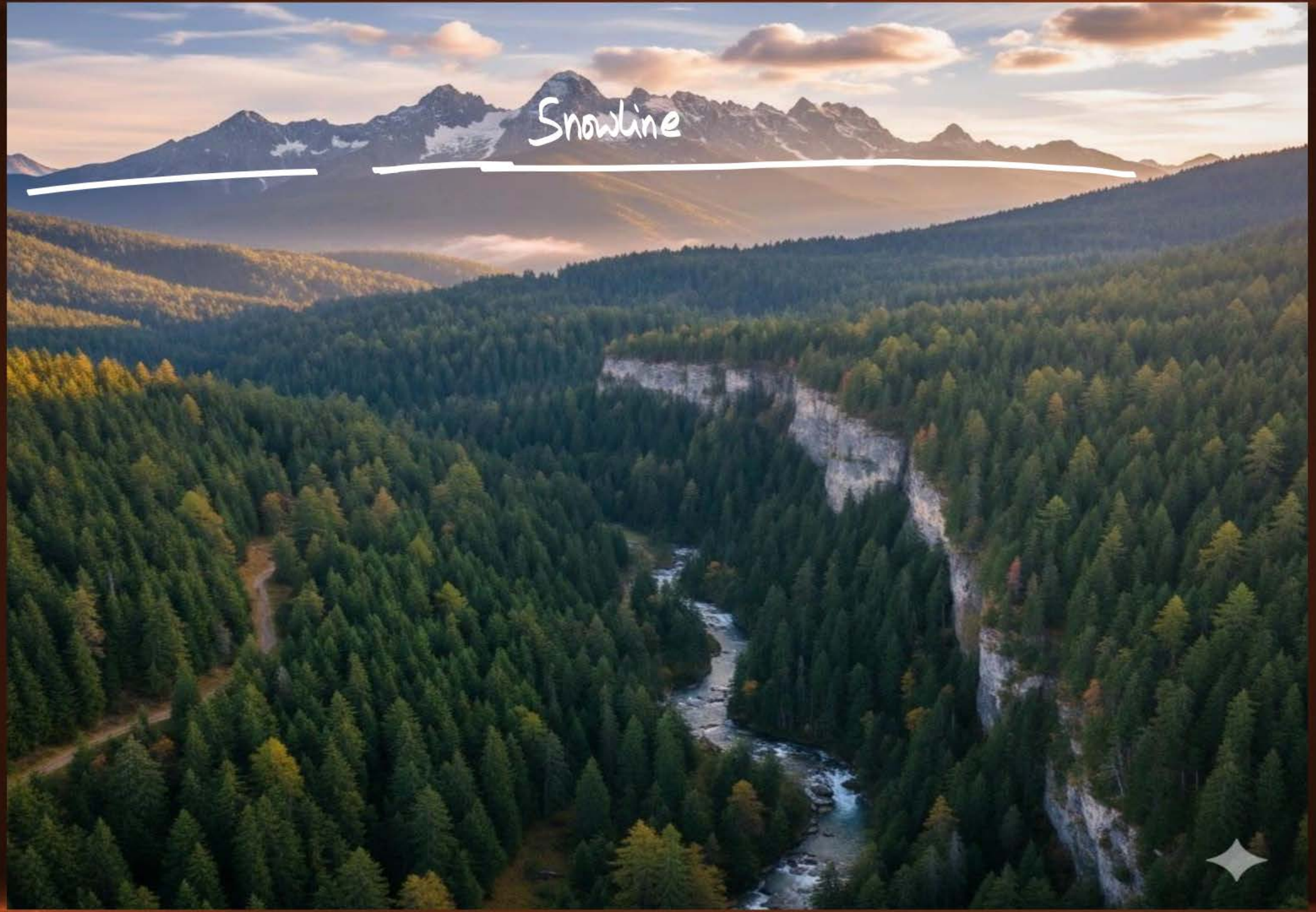
In the mountain regions temperature decreases with altitude. This has a corresponding effect on the natural vegetation of the mountain regions.

(a) Climatic Conditions:

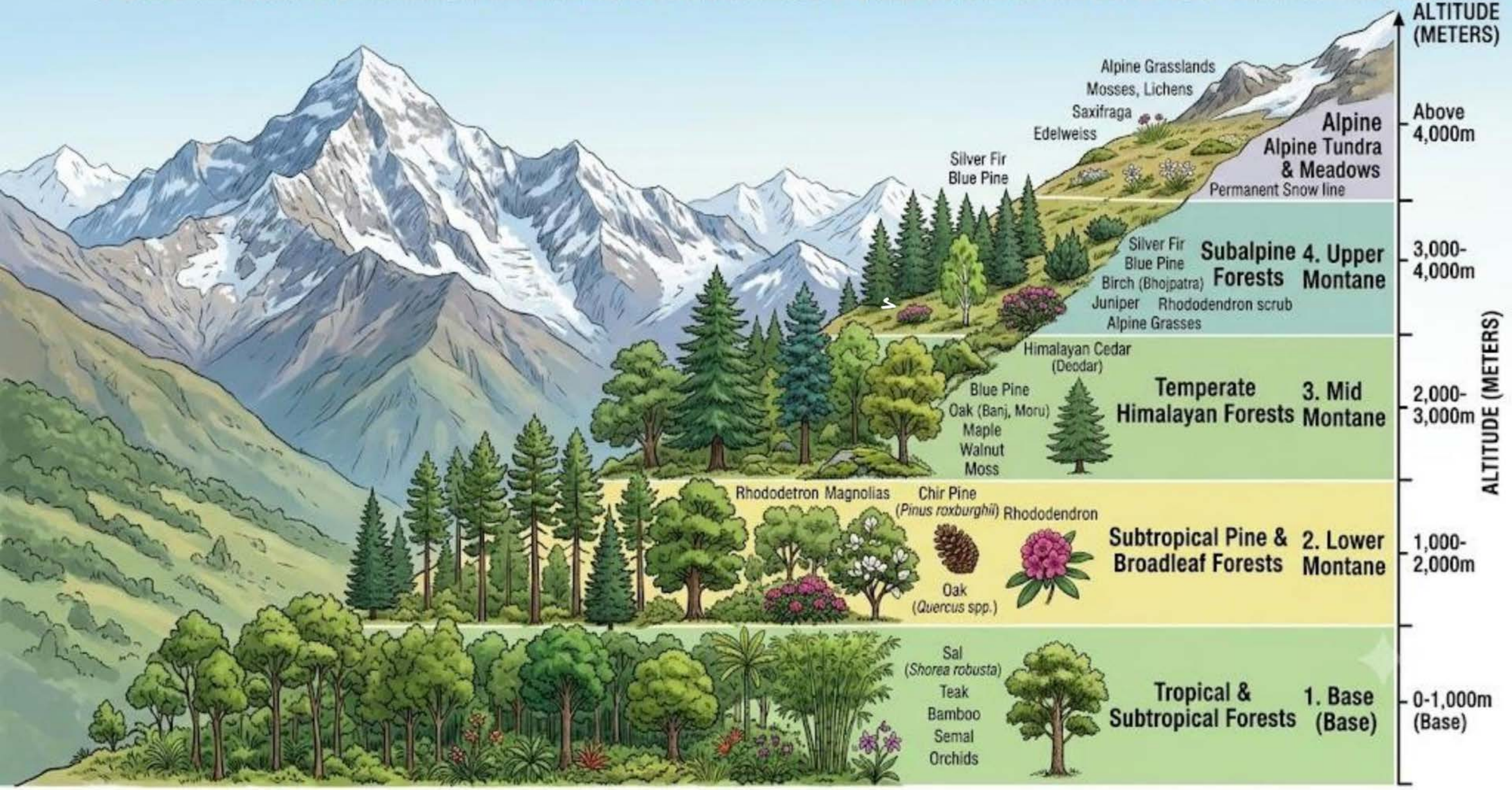
- annual temperature is 12° C to 13° C,
- rainfall is between 100 and 300 cm and
- annual humidity is between 56 and 65per cent.

(b) Relief: These forests occur at an altitude between 1000 m to 4000 m .

(c) Distribution: These forests cover the entire Himalayan zone. In the Peninsular region they are found in the Vindhyas, Nilgiris and the Western Ghats.


Snowline

VEGETATION ZONATION IN INDIAN MOUNTAIN FORESTS: ALTITUDINAL VARIATIONS



5. Mountain Forests



(d) Characteristic Features:

(i) These forests contain mixed species of broad leafed evergreen trees and conifers. They also contain scrubs, creepers and ferns.

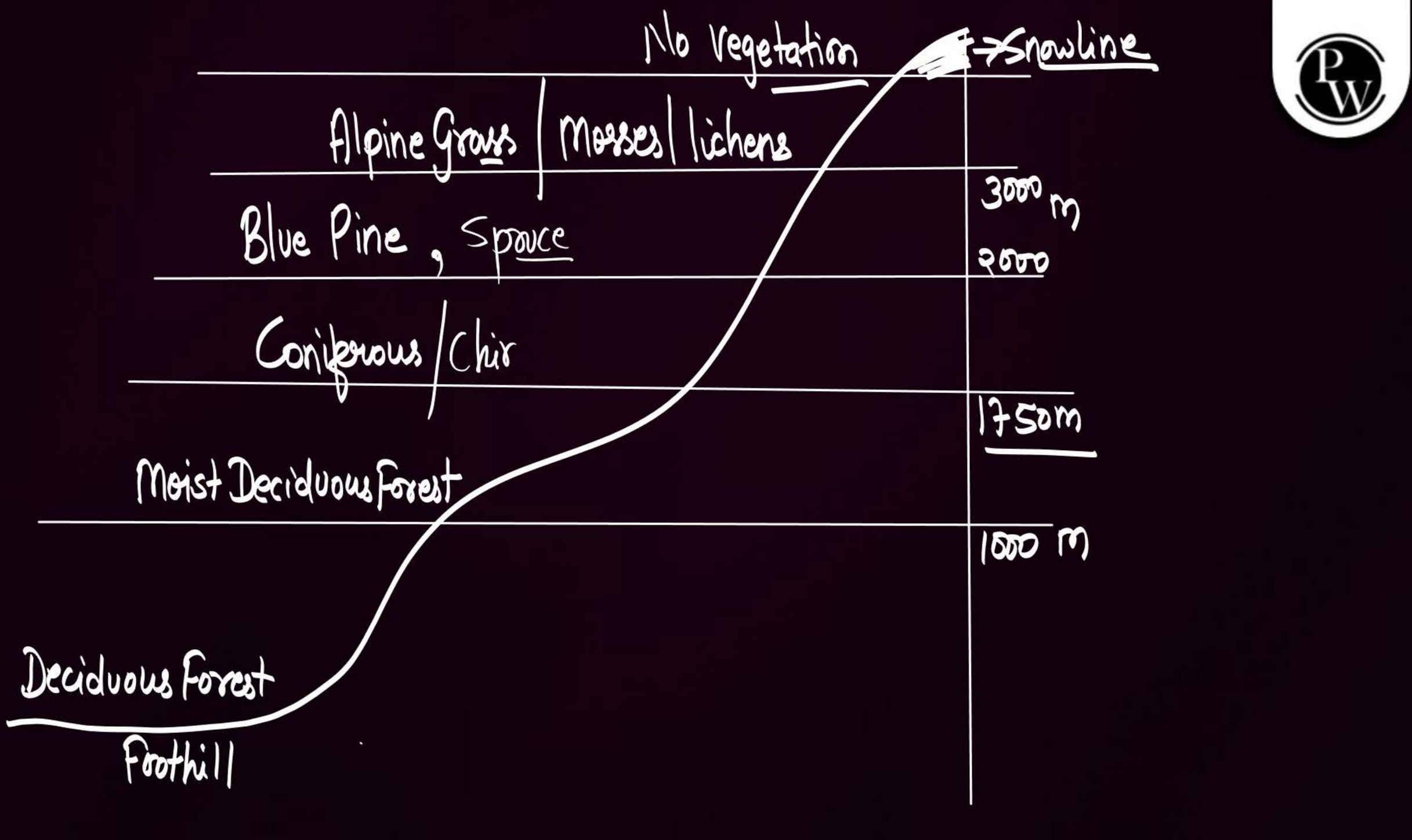
(ii) At the foothills of the Himalayas deciduous forests are found.

(iii) At an altitude of 1000 m to 2000 m they are the moist temperate forests.

(iv) In the hilly areas of West Bengal and Uttarakhand, evergreen broad leafed trees like chestnut and oak are found.

(v) Between the altitude of 1500 m and 1750 m , coniferous trees like Chir pine is an important tree.

(vi) At an altitude of 2250 m and 3000 m blue pine and spruce are found.

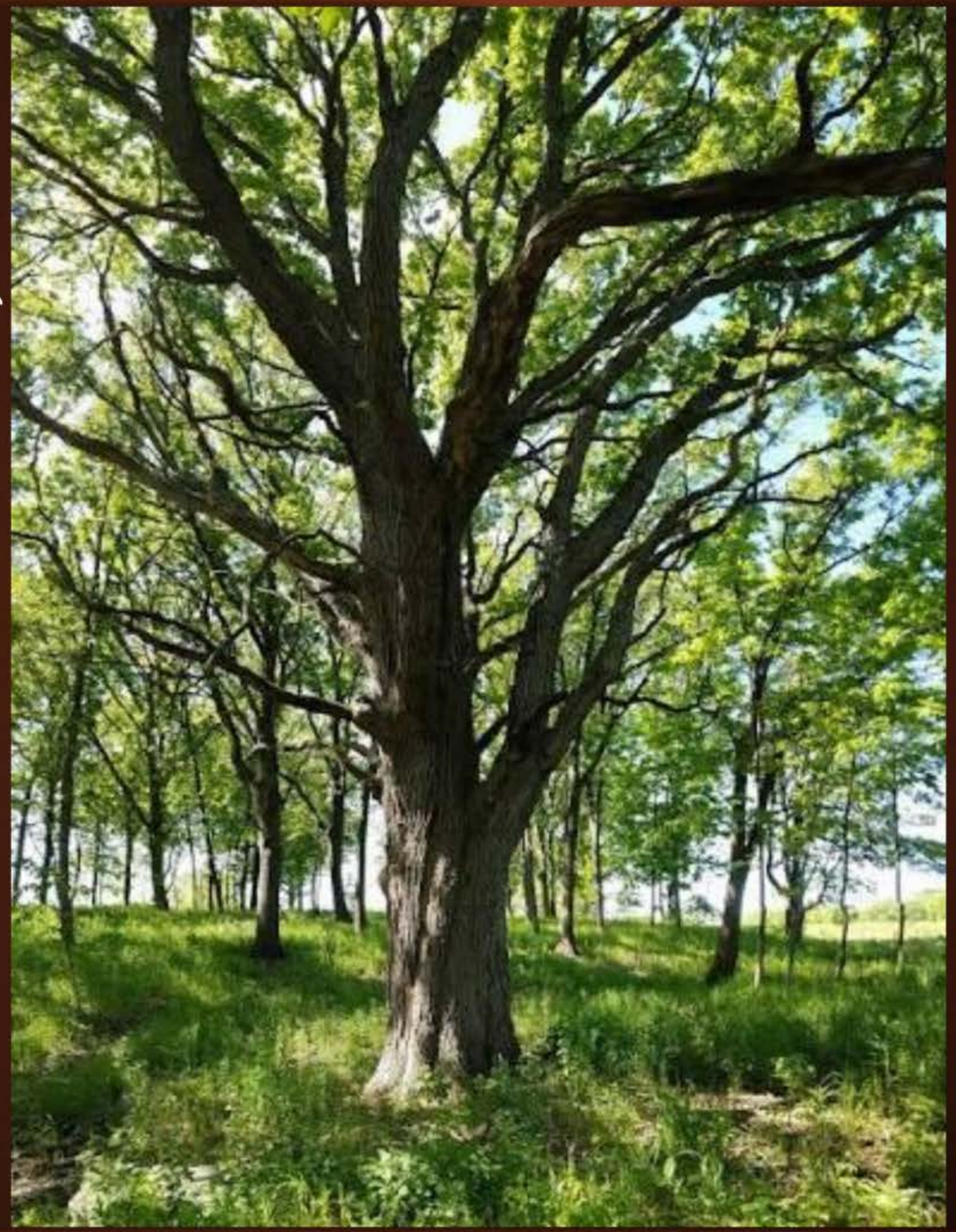

No Vegetation
Snowline
Alpine Grass | Mosses | lichens
Blue Pine , Spouce
Coniferous / Chir
Moist Deciduous Forest
Deciduous Forest
Foothill
3000 m
2000
1750m
1000 m


Conifers


Coniferous


Chestnut

Oak




Chirpine


Spruce


Blue Pine

5. Mountain Forests

(vi) At higher altitudes, alpine forests and alpine grass up to the snowline are found followed by mosses and lichens. Beyond the snowline no vegetation is found.

(vii) In the Peninsular regions, the area is only about 1500 m in height and vegetation varies from tropical to temperate forests.

- This is because of their closeness to the tropics and height of only 1500 m above sea level.
- So the vegetation is temperate in the higher regions, and sub tropical on the lower regions of the Western Ghats, especially in Kerala, Tamil Nadu and Karnataka.
- These temperate forests are called Sholas in the Nilgiris, Anamalai and Palani hills.


Alpine Grasses
Meadow
घास के मैदान


Mosses
lichens

5. Mountain Forests

(e) **Environmental Correlation:** These plants are adapted to conserve water and survive extreme conditions, preventing desertification.
The important trees found in these forests include magnolia, laurel, cinchona, wattle, plum, etc.



SELECTIVE TREES OF INDIA AND EXOTIC REGIONS

Vegetation Type	Climatic conditions	Distribution	Important Trees
Tropical Evergreen	Temp: 25° C-27° C Rainfall: More than 200 cm	Western slopes of the Western Ghats, hills of north-eastern region, and the Andaman and Nicobar Islands.	Rosewood, ebony, mahagony, toon chaplas, sissoo, gurjan, telsur, etc.
Tropical Deciduous	Temp: 24° C-27° C Rainfall: 100-200 cm	Northern states along the foothills of the Himalayas, eastern slopes of the Western Ghats.	Sal, teak, arjun, palas, shisham, mahua, mulberry, semul, sandalwood, Ber, etc.
Tropical Desert	Temp: 25° C-27° C Rainfall: Less than 50 cm	South-western Punjab, Haryana, Uttar Pradesh, Central and eastern Rajasthan, Madhya Pradesh and Gujarat.	Babool, date palm, khair, neem.
Littoral	Temp: 26° C-29° C Rainfall: Not more than 200 cm	Deltas of large rivers on the eastern coast, in pockets on the western coast, in saline swamps of the Sunderbans in West Bengal and coastal areas of Andhra Pradesh and Odisha.	Keora, amur, bhara, rhizophora, screw pines, canes and palms, sundari, agar, etc.
Mountain	Temp: 12° C-13° C Rainfall: 100-300 cm	The Himalayan zone.	Chestnut, oak, pine, spruce, magnolia, laurel, cinchona, wattle, plum, etc.

Forest Conservation

Man has been over-exploiting the forests to satisfy not only his needs but also his greed. This has led to a decline in the forest cover.

Reasons.

- Due to rapid population growth and the demand for more food, forests have been cleared to convert them into agricultural land.

घास के मैदान

- Forests have been converted into pasture land for expanding dairy farming and cattle ranching.

- Overgrazing in the forests by animals in the tropical and sub-tropical regions has resulted into large-scale degradation of natural vegetation.

Forest Conservation

- Ever increasing demand for timber (wood) for various purposes due to industrial expansion and urbanisation has added to the problem of deforestation.
- Construction of multi-purpose river valley projects has led to submergence of land and destruction of forested riversides.

Dams

Forest Conservation

All these factors have led to the decline in forests, which have the following effects:

(i) Decline in the forests have led to the decline in forest productivity.

(ii) Forests influence human environment by moderating climate, regulating water supply, maintaining fertility of soil and purifying air. Absence of forest cover leads to soil erosion which increases load of the rivers. The increased surface load reduces water in the rivers for human use.

Forest Conservation

(iii) Lack of forest cover reduces precipitation, thus causing droughts.

(iv) Forests act as a sink for carbon dioxide that the trees use during photosynthesis. The absence of forests increases the concentration of carbon dioxide in the atmosphere. Thus, increasing greenhouse effect in the atmosphere. This raises the temperature of the earth's surface and of the atmosphere leading to global warming and its harmful consequences.

Hence, conservation of forest is of vital importance for the survival and prosperity of human kind.



Leaves modified into spines/thorns is a characteristic feature of:

- A) Tidal vegetation
- B) Evergreen vegetation
- C) Temperate forests
- D) Xerophytic (Desert) vegetation

The Delta region of River Ganga and Brahmaputra is predominantly covered by:

- A) Tropical Thorn Forest
- B) Sunderbans (Tidal Forest)
- C) Coniferous Forest
- D) Tropical Dry Deciduous Forest

Tree leaves in Tropical Deciduous forests are shed for about 6 to 8 weeks in spring/early summer mainly to:

- A) Protect flowers from birds
- B) Prevent loss of moisture through transpiration
- C) Allow room for new parasitic plants
- D) Absorb more solar radiation

Coniferous

[ICSE Pattern] Why do trees in Montane (Mountain) forests have a conical shape at higher altitudes?

- A) To allow heavy snow to slide off easily without breaking branches
- B) To absorb maximum rain
- C) To prevent soil erosion
- D) To attract insects for pollination

Conservation Measures



To arrest deforestation and stop the diversion of forest lands for non-forest uses, the following conservation practices must be undertaken:

(i) **Increasing the Area Under Forests:** Loss of forests can be remedied by a massive programme of tree plantation.

Afforestation → Planting trees in barren or wasteland to increase forest cover.

Conservation Measures

Reforestration
↳ Planting trees in the areas where forest had been cut down.

(iii) **Stopping Indiscriminate Felling of Trees:** There should be a strict ban on felling of naturally growing trees. If a tree has to be cut, necessary permission has to be obtained. Besides, proper replacement of trees by planting saplings at least in the ratio of 1:10 (1 tree to 10 saplings) should be undertaken.

(iv) **Establishing Corridors between Different Reserved Forests:** Wide corridors should be established in different reserved forests to allow the migration of wild animals. This will also help numerous species of animals to use these corridors for their dispersal and migration.


Natural Corridors

Conservation Measures

(v) **Using Alternative Sources of Energy:** In many parts of the world including India, trees are felled for providing firewood. So, in order to conserve forests, we must use non-conventional or renewable sources of energy, like solar energy, tidal energy, hydel energy, etc.

(vi) **Proper Legislation and its Implementation:** Strict laws should be made to check deforestation. Proper care should be taken to see that these laws are strictly implemented.

Chipko movement
1970s

(vii) **People's Participation:** Participation of the local community is of utmost significance. Most programmes now involve local communities in planning, decision-making and implementation.

Conservation Measures

(viii) Developing badlands and barren lands into vegetation belts.

(ix) **Climate Action:** Reducing carbon footprints to mitigate climate change impacts and supporting global and national climate initiatives.

Protecting natural vegetation is crucial for maintaining biodiversity, preventing soil erosion, and ensuring ecological balance.


Saturated

Conservation Measures

Govt.

Wildlife Sancturies

National Parks.

Draft National Forest Policy

As a result of the implementation of National Forest Policy (1988) there has been an increase in forest and tree cover and reduction in the diversion of forest land for other purposes. Draft National Policy 2018 aims at sustainable forest management.

- The country should have a minimum of one-third of the total land area under forest and tree cover and two-thirds in the hills and mountainous regions. This will help to prevent soil erosion and land degradation and ensure the stability of the fragile eco-systems.

Social Forestry

- It refers to the management and protection of forests and afforestation on barren lands with the purpose of helping in the environmental, social and rural development.
- "forestry of the people, by the people, and for the people," .
- Social forestry is people-oriented, value based joint management of forests with the major objective of satisfying the needs, wants and aspirations of both the people and the government.
- Social Forestry is also known as extension forestry, mined forestry, urban forestry, tree farming, Van Mahotsav, recreation forestry, livestock forestry, etc.

Social Forestry

Objectives of Social Forestry

(i) Providing fuel wood, fodder for cattle, timber and minor forest products to rural people.

(ii) Utilising the available land according to its productive capacity.

(iii) Developing local cottage industries by providing raw materials.

(iv) Providing efficient conservation of soil and water.

(v) Providing employment opportunities to the rural people.

Social Forestry

(vi) Increasing agricultural production by using cow dung as manure.

(vii) Fulfilling the recreational needs of the people.

(viii) Improving the aesthetic scene of the area.

(ix) Achieving all-round rural development as a part of integrated rural development programme.

Agro-Forestry

Agriculture + Forestry

- Agro-forestry refers to the sustainable system of managing a piece of land through combined production of agricultural crops and forest crops and animal rearing, to ensure the most efficient land use under a management system in accordance with socio-cultural practices of the local people.

- It aims to provide conservation of the land to achieve a combined produce of forest and agricultural crops.



Objectives of Agro-Forestry

(i) To reduce pressure on natural forests for obtaining timber as well as non-timber forest produce.

(ii) To check soil erosion and to maintain the natural fertility of the soil.

(iii) To maintain ecological balance along with proper utilisation of farm resources.

(iv) To make the best use of all the available resources like land, manpower, livestock, ecological factors, etc., to obtain a variety of forest products such as food, fuel, fodder, livestock, recreation and a variety of forest products sustainably from the same land.

Objectives of Agro-Forestry

- Agro-Forestry differs from the traditional forestry as it is not vulnerable to population stresses.
- It is created in accordance with the socio-cultural practices of the population.
- In fact, agro-forestry co-ordinates with local population and reduces the pressure of population on traditional forestry.

Sustainable Agro-Forestry to Combat Hunger in India

- One Tree Planted a non-profit tree planting charity that plants trees in countries around the world has partnered with Sustainable Green Initiative (SGI), a local organisation with one objective: to plant (mostly) fruit trees to fight hunger, poverty and climate change.
- Under this model, fruit tree saplings are distributed for 1-3 years in community lands, homesteads of marginal farmers, government school campuses, orphanages, old age homes etc.
- Trees planted include lemon, guava, custard apple, gooseberry, pomegranate, jack fruit, wood apple and tamarind.

Miyawaki Method

The Miyawaki method is a technique for creating dense native forests in a short period. Named after its developer Akira Miyawaki, a Japanese botanist, it has been successfully implemented in urban areas all over the world.

Van Mahotsav

- Van Mahotsav, literally meaning "Forest Festival", is an annual tree-planting festival celebrated during the first week of July.
- The first tree plantation week was organized by M.S. Randhawa, a botanist from Punjab in 1947 from July 20 to 27 in Delhi.
  - ↳ Those who study plants
- Prominent leaders like Jawaharlal Nehru and Dr. Rajendra Prasad attended this event.

Van Mahotsav

- Dr. K.M. Munshi, the Union Minister for Agriculture and Food, officially launched Van Mahotsav as a national event.
- He moved the celebration to the first week of July to coincide with the monsoon season, which is ideal for planting trees.
- Van Mahotsav has since grown into a significant environmental campaign, with millions of trees planted across India each year.

Van Mahotsav

The objective of this festival includes:

- Raising awareness about the importance of trees;
- increase in fruit production;
- creation of shelter-belts across agricultural fields;
- providing fodder for cattle;
- prevention of soil infertility;
- encouraging soil conservation; and
- enabling supply of timber for various uses.

Social Forestry involves forest management and tree planting on barren lands with the involvement of:

- A) Foreign timber companies
- B) Local communities and rural people
- C) Only military forces
- D) Urban industrial developers

What is the main objective of Agroforestry?

- A) Combining agriculture with tree planting on the same farmland
- B) Clearing forests for commercial palm cultivation
- C) Burning forest land for shifting cultivation
- D) Converting national parks into farmlands

2024

[ICSE Pattern] Match the following trees with their forest types:

- *Trees*: 1. Rosewood, 2. Teak, 3. Babool, 4. Sundari
- *Types*: i. Tidal, ii. Evergreen, iii. Thorn, iv. Deciduous
- A) 1-ii, 2-iv, 3-iii, 4-i
- B) 1-iv, 2-ii, 3-i, 4-iii
- C) 1-i, 2-iii, 3-ii, 4-iv
- D) 1-iii, 2-i, 3-iv, 4-ii

[ICSE Pattern] Van Mahotsav is an annual program in India associated with:

- A) Cutting old trees for lumber
- B) Tree-planting festival to increase forest cover
- C) Dam construction in forest rivers
- D) Clearing land for cattle grazing

[ICSE Pattern] Afforestation refers to:

- A) Cutting trees for industrial fuel
- B) Natural regrowth without human interference
- C) Planting trees on land that previously had no forest cover
- D) Clearing forests for urban expansion

[ICSE Pattern] Which of the following is a key measure taken by the government for forest conservation?

- A) Encouraging shifting cultivation (Jhum)
- B) Promoting unrestricted commercial logging
- C) Establishing National Parks and Wildlife Sanctuaries
- D) Converting forest land into tea estates without regulation

[ICSE Pattern] "Re-planting trees in a forest area where original trees were destroyed or removed" is known as:

- A) Deforestation
- B) Overgrazing
- C) Reforestation
- D) Commercial harvesting

PYQ

i) Mention the climatic conditions that lead to Tropical Desert Vegetation?

ii) Why do mountains have varied vegitation?

iii) Why it it important to grow forest in the area where Iron & Steel industry is located?

iv) Suggest any two ways for forest Conservation?

Thank You